x	y
1	2,5
2	4.3
3	5.5
4	6.1
5	6.1

Q.1)

a) $\hat{y} = w_1 x + w_2 x^2$

$X.w = y$ formatını yakalamamız gerekiyor.

w_1'in yanında x w_2'nin yanında x^2 var.

$$X = \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \\ 4 & 16 \\ 5 & 25 \end{bmatrix}, \quad y = \begin{bmatrix} 2,5 \\ 4.3 \\ 5.5 \\ 6.1 \\ 6.1 \end{bmatrix}, \quad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

b) find $w^* = (X^T.X)^{-1}.X^T.y$.

$$X^T = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 9 & 16 & 25 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \\ 4 & 16 \\ 5 & 25 \end{bmatrix}$$

Bu tabloyu uygula

1.satır x 1.sütun

$(1.1) + (2.2) + (3.3) + (4.4) + (5.5) = 55$

1.satır x 2.sütun

$(1.1) + (2.4) + (3.9) + (4.16) + (5.25) = 225$

2.satır x 1.sütun

$(1x.1) + (4.2) + (9.3) + (16.4) + 25.5 = 225$

2.satır x 2.sütun

$(1x1). + (4.4) + (9.9) + (16.16) + (25.25) = 979$

$$X^T.X = \begin{bmatrix} 55 & 225 \\ 225 & 979 \end{bmatrix}$$

$$X^T.y \Rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 9 & 16 & 25 \end{bmatrix} \begin{bmatrix} 2,5 \\ 4,3 \\ 5,5 \\ 6,1 \\ 6,1 \end{bmatrix}$$

1.satır x sütun

$(1 . 2,5) + (2. 4,3) + (3. 5,5) + (4 . 6,1) + (5. 6,1) = 82,5$

2.satır x sütun

$(1. 2,5) + (4. 4,3) + (9. 5,5) + (16. 6,1) + (25. 6,1) = 319,3$

$$X^T.y = \begin{bmatrix} 82,5 \\ 319,3 \end{bmatrix}$$

$$[X^T.X]^{-1} \Rightarrow \underbrace{\begin{bmatrix} 55 & 225 \\ 225 & 979 \end{bmatrix}}_{A} \Rightarrow \begin{bmatrix} 979 & -225 \\ -225 & 55 \end{bmatrix} \cdot \frac{1}{\underset{3220}{Det(A)}} = A^{-1}$$

$Det(A) = 55.979 - (225.225) = 3220$

$$\underbrace{\frac{1}{3220} \cdot \begin{bmatrix} 0,304 & -0,069 \\ -0,069 & 0,017 \end{bmatrix}}_{[X^T.X]^{-1}} \cdot \underbrace{\begin{bmatrix} 82,5 \\ 319,3 \end{bmatrix}}_{X^T.y}$$

$$= \begin{bmatrix} 8925 \\ -1001 \end{bmatrix} \cdot \frac{1}{3220} = \underbrace{\begin{bmatrix} 2,7717 \\ -0,31 \end{bmatrix}}_{\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}}$$

$$\boxed{\hat{y} = 2,77x - 0,31x^2}$$

c) Determinantı 0'dan farklı old. için o sistemin matematiksel olarak 1 çözümü vardır.

This model is unique. Because there is one solution.

d) Hayır ridge regression'a gerek yoktur. Çünkü matrisin det 0 çıkmadı ve determinantını alabildik. Hata yüzeyimizin tek bir minimum nok. var, sistem zaten en iyi sonucu buldu, ekstra bir düzeltmeye (ridge regression'a) gerek yok.

Question 2

$$\hat{Y} = \begin{cases} 1, & w_1 x_1 + w_2 x_2 + w_0 \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

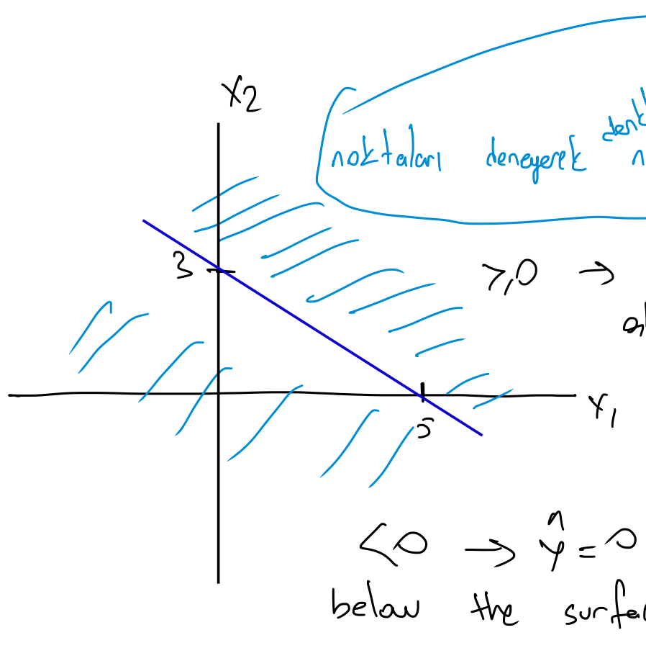
sınırı belirleyen çizgi =0 çizgisi

$w_1 = 3$, $w_2 = 5$, $w_0 = -15$

$3x_1 + 5x_2 - 15 = 0$

$x_1 = 0$ $\quad 0 + 5x_2 = 15$ $\quad \underline{x_2 = 3}$

$x_2 = 0$ $\quad 3x_1 = 15$ $\quad \underline{x_1 = 5}$



noktaları deneyerek denkleme koyarak neresi "−" neresi "+" bulursun

$\geq 0 \rightarrow \hat{y} = 1$ above the surface

$< 0 \rightarrow \hat{y} = 0$ below the surface

b) Olasılık olarak düşünüyoruz.

$$P = [Y=1] = \frac{1}{1 + e^{-[w_1 x_1 + w_2 x_2 + w_0]}}$$

$x_1 = 3$ $\quad x_2 = 9$ $\quad w_1 = 3$, $w_2 = 5$, $w_0 = -15$

$$= \frac{1}{1 + e^{-[9 + 45 - 15]}}$$

$$P = [Y=1] = \frac{1}{1 + \underbrace{e^{-39}}_{0}} > 0,5 \text{ (olasılığımız)}$$

Probability?

$$\frac{1}{1+0} \cong 1 \quad 0,5\text{'den kesin büyük}$$

$P=[Y=1]$'in olasılığımız 0,5'den büyük çıktığı için sistemin tahmini $\hat{Y} = 1$ olacaktır.

Q.4) Decision tree

X_1	X_2	Y
1	1	1
1	1	1
1	1	2
1	0	3
0	0	2
0	0	3

a) Which feature X_1, X_2 provides more information gain? - - -

$X_1 = 1 \Rightarrow Y = (1, 1, 2, 3)$ → çeşitlilik çok

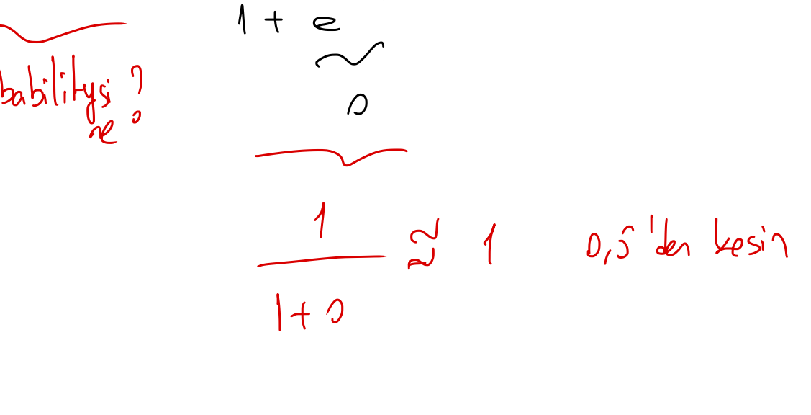
$X_1 = 0 \Rightarrow Y = (2,3)$ → çeşitlilik az, ancak ağırlıklı olan yok.

$X_2 = 1 \Rightarrow Y = (1, 1, 2)$ → ağırlıklı 1 Bunu tercih ederiz.

$X_2 = 0 \Rightarrow Y = (2,3,3)$ → ağırlıklı 3

X_2 provides more information gain.

b)

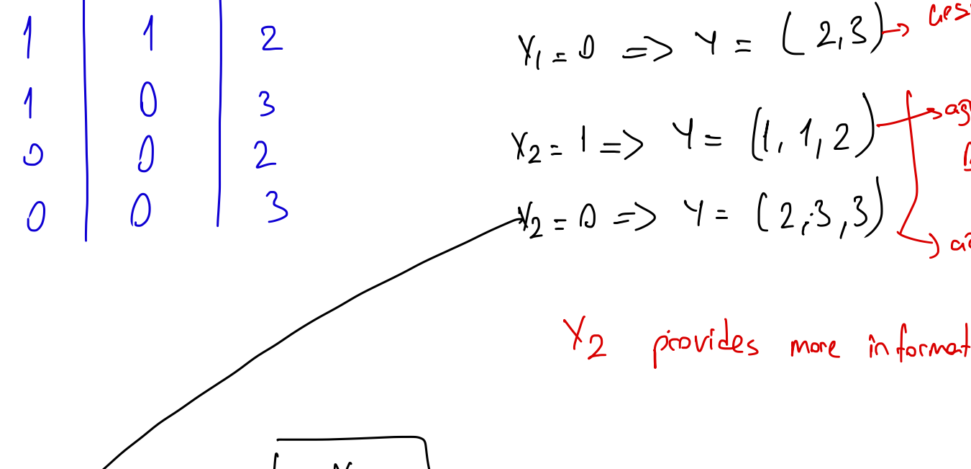

X2
X2=0
X2=1
Y=3
Y=1
3 ağırlıklıydı o yüzden tahmin ederek 3 yazdık
1 ağırlıklıydı " " 1 yazdık

c) How would this tree classify an example with $X_1=0$ and $X_2=1$?

$X_1 = 0$ önemsiz $\quad \underline{X_2 = 1}$ bakarız (ağacımız X_2 sorar)

$\underline{Y = 1}$

2024 – MT



1. Consider the following data set with three binary features X_1, X_2, X_3, and a binary label Y.

X_1	X_2	X_3	Y
1	0	1	0
0	0	1	0
1	1	0	0
0	0	0	1
1	1	1	1
1	0	0	1

(a) [10 pt] We wish to fit a Naive Bayes model to the data $P(Y|X_1,\ldots,X_n) \propto P(Y)\prod_{i=1}^{n} P(X_i|Y)$. Calculate the probability $P[Y = 0|X = (0,1,1)]$. =>

(b) [15 pt] Apply Laplace smoothing with $k = 1$. Calculate $P[Y = 1|X = (0,1,0)]$.

a) $P[Y=0|X_1=0, X_2=1, X_3=1] = P[Y=0] . P[X_1=0|Y=0] . P[X_2=1|Y=0] . P[X_3=1|Y=0]$

$$= \frac{3}{6} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} = 1/27$$

b) Apply laplace smoothing with $k=1$. Calc. $P[Y=1|X=(0,1,0)]$

1 tane kopya yazacaksın (k=2 olsa alt kısma 2 tane kopyalardık)

eksiği bulup ona göre ekleniyor 6 satır var zaten tabloda. $2^3 = 8$ binary satırı olmalıydı. Demekki 2 tane eklememiz lazım.

X_1	X_2	X_3	Y	
1	0	1	0	given
0	0	1	0	
1	1	0	0	
0	1	1	0	added
0	1	0	0	
0	0	0	1	given
1	1	1	1	
1	0	0	1	
0	1	1	1	(copy) added
0	1	0	1	

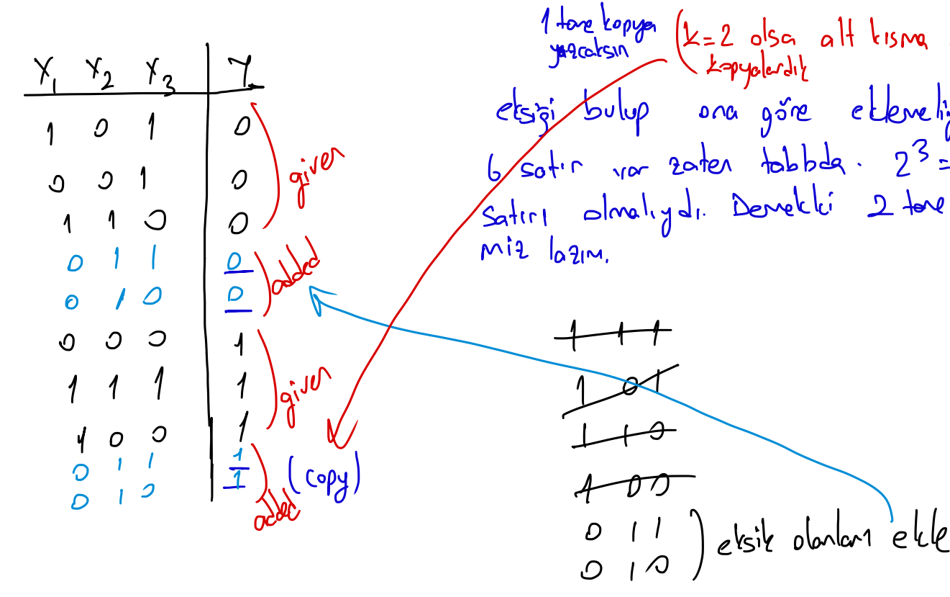
~~1 1 1~~
~~1 0 1~~
~~1 1 0~~
~~1 0 0~~
0 1 1) eksik olanları ekle
0 1 0
~~0 0 1~~
~~0 0 0~~

Soru: $P[Y=1|X=0,1,0)]$

$$= P[Y=1] . P[X_1=0|Y=1] . P[X_2=1|Y=1] . P[X_3=0|Y=1]$$

$$= \frac{1}{2} \cdot \frac{3}{5} \cdot \frac{3}{5} \cdot \frac{3}{5}$$

$$= \frac{27}{250}$$



2. [25 pt] Suppose you are an avid BTS (a South Korean boy band that formed in 2010 and debuted in 2013) stan who monitors the social media accounts of each of the members. Suppose you wish to find the probability that a BTS member will post at any time of day. Over the three days period, you look on Instagram and find the following number of new posts: x = [3,4,1]. A fellow stan tells you that this comes from a Poisson distribution.

$$p(x|\theta) = \frac{e^{-\theta}\theta^x}{x!}$$

where θ is an unknown parameter that we want to estimate. You are told that θ has the following probability density function (PDF)

$$p(\theta) = \frac{1}{4}\theta e^{-\frac{\theta}{2}}, \; \theta > 0$$

Calculate maximum a posteriori probability (MAP) estimate $\hat{\theta}_{MAP}$. MAP kuralı = ikisini çarp

$x = 1 \Rightarrow P(1|\theta) = \frac{e^{-\theta}\theta^1}{1!}$

$x = 3 \Rightarrow P(3|\theta) = \frac{e^{-\theta}\theta^3}{3!}$

$x = 4 \rightarrow P(4|\theta) = \frac{e^{-\theta}.\theta^4}{4!}$

olasılıklar birbiriyle çarpılır.

$$P(x|\theta) \quad e^{-\theta}.\theta . \frac{e^{-\theta}.\theta^3}{3!} . \frac{e^{-\theta}.\theta^4}{4!} = \frac{e^{-3\theta} . \theta^8}{4!.3!}$$

$$P(\theta) = \frac{1}{4}\theta e^{\frac{-\theta}{2}}, \quad \theta > 0$$

$$\frac{1}{4} . \theta . e^{-\theta/2} . \frac{e^{-3\theta}.\theta^8}{4!.3!} = \frac{\theta^9 . e^{-7\theta/2}}{4!.4!}$$

$$\log\left(\frac{\theta^9}{4!.4!}\right) + \log\left(e^{-7\theta/2}\right)$$

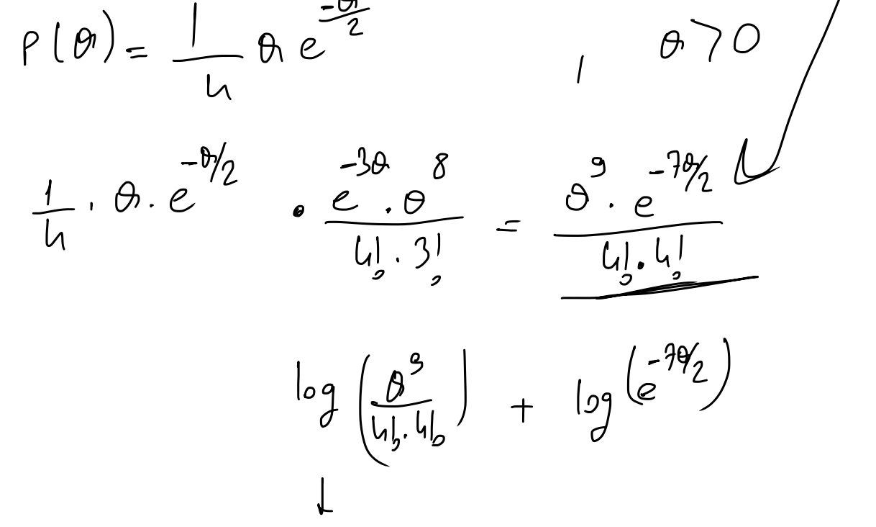
$$= \underbrace{\log(\theta^9)}_{ln} - \underbrace{\log(4!.4!)}_{ln} + \underbrace{\log e^{(-3,5\theta)}}_{ln}$$

$$= 9\ln(\theta) - \ln(4!.4!) - 3,5\theta . \underset{1}{\ln(e)}$$

$$= 9\ln(\theta) - 6,36 - 3,5\theta$$

find maximum a posteirori prob. (MAP) estimate $\hat{\theta}_{MAP}$

$$\frac{9}{\theta} - 3,5 = 0 \qquad \underline{\theta = 2,5714}$$



3. In this problem, you will see how to learn a decision tree. Consider the following data set with three features X_1, X_2, X_3, and label Y.
   (a) [10 pt] Suppose calculate the mutual information for each feature (X_1, X_2, X_3) to determine the root node for a Decision Tree trained on the data. What feature should be split on at the root node?
   (b) [10 pt] Calculate what the next split should be.
   (c) [5 pt] Draw the resulting tree.

Outlook (X_1)	Temperature (X_2)	Humidity (X_3)	Play Tennis? (Y)
sunny	hot	high	no
overcast	hot	high	yes
rain	mild	high	yes
rain	cool 0	normal	yes
sunny	mild	high	no
sunny	mild	normal	yes
rain	mild	normal	yes
overcast	hot	normal	yes

$H(y)$ = karmaşa seviyesi (Entropi) $\qquad H(y) = -p\log_2(p)$ → (p) olasılıktır.

$I(y;x)$ = mutual inform. (bilgi kazancı)

Y: 6 yes, 2 No

$$H(y) = -\frac{6}{8}\log_2\left(\frac{6}{8}\right) - \frac{2}{8}\log_2\left(\frac{2}{8}\right) \cong 0,811$$

X_1:

Sunny: 1 YES 2 NO ⇒ Entropi $= -\frac{1}{3}.\log_2\left(\frac{1}{3}\right) - \frac{2}{3}.\log_2\left(\frac{2}{3}\right) \cong 0,918$

Rain: 3 Yes = Entropi = 0

overcast: 2 Yes ⇒ Entropi = 0

ortalama: $\left(0,918 \times \frac{\overset{(adet)}{3}}{8}\right) + 0 \times \frac{\overset{adet}{(3)}}{8} + 0 \times \frac{\overset{(adet)}{2}}{8} = 0,344$

$I(Y;X_1) = 0,811 - 0,344 = 0,467$

X_2:

hot: 2Y 1N → $-\frac{2}{3}.\log_2\left(\frac{2}{3}\right) - \frac{1}{3}.\log_2\left(\frac{1}{3}\right) \cong 0,918$

mild: 3Y 1N → $-\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}.\log_2\left(\frac{1}{4}\right) \cong 0,811$

cool: 1Y → 0

for X_2 = $\left(0,918 \times \frac{3}{8}\right) + \left(0,811 . \frac{4}{8}\right) + \left(0 . \frac{1}{8}\right) \cong 0,75$

$I(Y;X_2) = 0,811 - 0,75 = 0,061$

for X_3 ⇒ high = 2Y 2N ⇒ $-\frac{1}{2}.\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2(1/2) \cong 0,5$

normal = 0 $\qquad I(Y;X_3) = 0,811 - 0,5 = 0,311$

I'sı en yüksek yani en fazla karmaşayı düşüren X_1

X_1 should be

b) Calculate the next split should be.

X_1'i ağacımızın tepesine koyduk.

X_1 = Sunny: 1 Y 2 N = →

Rain: 3 Y = 0

Overc: 2 Y = 0

X_2	X_3	Y
hot	high	no
mild	"	no
"	normal	yes

→ entropisi $H(Y) = 0,918$

X_2 sorarız.

hot 1 No = entropi = 0

mild 1N 1Y $= -\frac{1}{2}.\log_2 1/2 - \frac{1}{2}\log_2 1/2 = 1$

ort. $1.\frac{2}{3} + \frac{1}{3}.0 = \frac{2}{3} = 0,67$

$I(Y;X_2) = 0,918 - 0,67 = 0,248$

X_3'ü sorarız. (entropi)

high 2 No = 0

normal 1 Yes = 0

$H(Y,X_3) = 0$

$I(Y;X_3) = 0,918 - 0 = 0,918$

We choose X_3

c) Draw tree